CANALRUNG · CALCULATION

CanalRung sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-11-05	Amish Chadha	Kit 1.9.1: requirement identifiers R1 to R10 mapped to PR-001 to PR-013 of CNR-PRS-001; no calculation changed
0.3	2026-10-03	Amish Chadha	Amish's requirement decisions of 2026-10-03 (4A, 9A; CNR-DDR-003): 9 mm EN 1891 type B ropes (E1, B1 to B3, C1 to C3); R2 measured against the restated 4.5 kg; the 500 x 1,200 mm footing standard at every site (G4 to G6); cost recomputed (I1, I2)
0.2	2026-10-03	Amish Chadha	Design for construction (CNR-DDR-002) applied: rope-through-rung joint, stand-off blocks, crush tube, rope cut length and mass, cost by build-order BOM
0.1	2026-10-03	Amish Chadha	First issue for TRL 3: reach, mass, buoyancy, rung and rope strength, loads in use, anchor and footing, deployment, cost

CanalRung sizing calculations

On paper the constructable CanalRung meets nine of its ten requirements and is over its value-engineering target on the tenth. The 4,020 mm rung section puts two rungs in the water on the design slope (1.5:1, water 2.0 m below the coping) and four on a 3.0 m vertical wall. Every rung floats with 0.22 to 0.28 kg to spare and the bottom three carry the soft throw weight 1.75 times over. A rung under a 1.5 kN point load reaches 0.56 of yield, and the two 9 mm ropes with their stopper knots hold 18.4 kN wet, 4.08 times the 4.5 kN target. The anchor holds 4.5 kN with factors of 1.53 (eye nut) and, for the 500 x 1,200 mm footing now standard at every site, 3.58 in drained soil and 1.59 in saturated soil. The ladder as thrown weighs 4.27 kg against the 4.5 kg target Amish restated on 2026-10-03 (R2 met on mass), and the parts cost USD 605.20 against the USD 500 value-engineering target. Every number below is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C2], is the line of that script's output that carries it.

> **Safety:** These are first-principles estimates for a paper proof of concept. They do not show that the ladder, its knots or the anchor are safe to use. Rope, knot, rung and anchor strengths must be proof-loaded (CalRig, TRL 4) and the climb, throw and deployment tried in a supervised test tank before any station is placed at a canal. Never enter the water to rescue someone. See CNR-PRC-001, Safety.

Scope and method

The note checks every requirement in CNR-REQ-001 against the constructable design in `cad/src/model.py` (CNR-DDR-002). The script imports the model's parameters and builds the ladder once for part volumes, so the rungs, float sleeves, bushes and stand-off blocks used here are the ones in the STEP files and drawings. It reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`.

The design case is the station 1,000 mm back from the coping of a concrete-lined canal with a 1.5:1 (horizontal to vertical) side slope, the water 2.0 m below the coping and flowing at up to 2.0 m/s, and a 100 kg adult climbing out.

Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Climber	100 kg, clothed and wet; dynamic factor 1.5 for grabbing and climbing (1.47 kN)	Screening value for a heavy adult
Rung	6061-T6 drawn tube, 240 MPa minimum yield; load as a point at mid-span between the ropes (320 mm), simply supported	Conservative: hands and feet usually load nearer the ends
Rope	9 mm EN 1891 type B, minimum breaking strength 18 kN, about 55 g/m; overhand stopper knot keeps 60 % (literature 55 to 65 %); wet 85 %; over a 10 mm unprotected edge 50 %	Typical published values; to be proof-loaded
Buoyancy	Fresh water 1,000 kg/m3; foam 33 kg/m3 closed cell; aluminium 2,700, nylon 1,140, HDPE 950, polyester 1,380 kg/m3; tube bore floods through the rope holes; rope share per rung includes 100 mm in each knot	Handbook densities
Current	2.0 m/s surface current; person holding a rung, drag area 0.35 m2; rung drag coefficient 1.2	Screening values
Anchor	Eye nut working load limit 700 kg in line with the thread (DIN 582 class, M16); 10 mm screw link 1,000 kg; M16 grade 8.8 proof 600 MPa on 157 mm2	Catalogue classes; the bought parts must be marked
Post	S235 square hollow section 60 x 60 x 3; anchor 250 mm above ground	
Soil	Drained, medium dense granular bank: unit weight 18 kN/m3, friction angle 30°; saturated case 8 kN/m3 submerged	Screening values; to be confirmed on site
Footing	Short rigid pile in cohesionless soil (Broms), free head: ultimate lateral load 0.5 γ D L^3 Kp / (e + L)	Broms (1964)

A. Reach and climbing geometry (R1, R5)

The rung section is 4,020 mm: 13 rungs at 335 mm pitch [A1]. On the 1.5:1 design slope (33.7°), rung 1 lies 150 mm down the wall from the coping [A2] and the bottom rung ends 2,313 mm below the coping, 313 mm below the water line, so rungs 12 and 13 are in the water and rung 11 is 59 mm above it [A3]. Because the rungs float, the ones that reach the water rise to the surface. The ladder still puts two rungs at or below the water line when the water is 2,127 mm below the coping [A4]. On a vertical wall with the water 3,000 mm down, the bottom rung is 1,170 mm under water with four rungs in it, and two rungs reach the water down to 3,835 mm [A5]. R1 is met for the stated design envelope; deeper sloped canals need the longer variant.

The clear width between the rope bushes is 298 mm, the pitch 335 mm, and at each stand-off rung the float sleeve stands 70 mm off the wall, room for a hand [A6]. R5's geometry is met by design; the climb time needs a tank trial.

B. Mass (R2)

Table 2. Mass of the ladder as thrown [B2].

Part	Mass (kg)
Rungs (13)	1.715
Float sleeves (13)	0.161
Rope bushes (26)	0.091
End caps (26)	0.104
Stand-off blocks (8)	0.420
Stand-off bolts (8)	0.080
Side ropes, 9 mm, 2 x 7.95 m at 55 g/m [B1]	0.874
Seizings	0.020
Screw links (2)	0.120
Throw weight with pouch	0.530
Edge protectors (2)	0.160
Total	**4.27**

The ladder as thrown weighs 4.27 kg, 0.23 kg inside the 4.5 kg target of R2 [B3]. The 9 mm ropes save 0.21 kg against the 10.5 mm ropes they replace, which was not enough to reach the earlier 4.0 kg (still 0.27 kg over), so R2 was restated to 4.5 kg under Amish's fallback in decision 4A (CNR-DDR-003). **R2 is met on paper on mass**; throw accuracy needs trials. Rungs of 25.4 x 1.65 mm tube would save 0.20 kg [B4] but would take the rung past the stress limit (D2), so they were rejected.

C. Buoyancy (R3)

A plain rung with its share of rope floats with 0.277 kg to spare and a stand-off rung with 0.221 kg, both above the 0.15 kg target [C1]. In water the throw weight weighs 0.442 kg; rungs 11, 12 and 13 together lift 0.775 kg, a factor of 1.75 against the 1.5 target [C2], so the weight hangs below the floating rungs instead of pulling them under. The whole ladder in water has 2.93 kg of spare lift [C3]. The lighter rope gives slightly more spare lift than before. **R3 is met on paper.** The tube bore is not counted: it floods through the rope holes.

D. Rung strength (R4)

The 28.6 x 1.65 mm 6061-T6 tube has a section modulus of 890 mm3. A 1.5 kN point load at mid-span over the 320 mm between the ropes gives 120 N m and 135 MPa, 0.56 of the 240 MPa minimum yield, inside the two-thirds limit [D1]. The lighter 25.4 x 1.65 mm tube would reach 175 MPa, 0.73 of yield [D2].

Each rung end passes 750 N through the bush flange onto the knot; the nylon flange bears at about 17 MPa on a 2 mm contact band, against about 50 MPa for nylon to yield [D3]. With the whole climber on one stand-off rung on the 1.5:1 slope, each block pushes on the wall with 612 N; wall friction tries to turn the block round the rung with 17.4 N m, which puts 610 N across the M5 bolt, against about 6 kN for its two shear planes [D4].

E. Ropes, knots and links (R4)

One 9 mm side rope with a stopper knot, wet, holds 18 x 0.6 x 0.85 = 9.18 kN; both ropes 18.4 kN, 4.08 times the 4.5 kN target [E1]. That is over the 18 kN (four times) that R4 asks for, and well over the floor of three times (13.5 kN) that Amish set for the smaller rope in decision 4A. The margin over R4 is thin, so the bought rope must carry a declared breaking strength of at least 18 kN and the proof load at TRL 4 must include knotted, wet rope. If a rope were pulled over an unprotected 10 mm coping edge as well, both ropes would still hold 9.2 kN [E2]; the edge protectors are there to keep the rope off that edge. The 10 mm screw links have a working load limit of 9.8 kN, 2.18 times the target [E3]. **R4 is met on paper.**

F. Loads in use

The design climber with the dynamic factor loads the ladder with 1.47 kN [F1]. On the 1.5:1 slope the ropes carry only the part of that along the wall, 0.82 kN (wall friction ignored); on a vertical wall they carry all 1.47 kN [F2]. A person holding a rung in a 2.0 m/s current adds 700 N of drag, and three rungs in the water 106 N [F3]. Adding the vertical-wall climb and the full current drag together, which cannot both happen at full value, gives an upper bound of 2.32 kN on the anchor, inside the 4.5 kN target by a factor of 1.94 [F4].

G. Anchor: eye nut, bolt, post and footing (R6)

Table 3. Anchor elements against the 4.5 kN target.

Element	Capacity	Factor	Tag
Forged M16 eye nut, in line	6.87 kN working load limit	1.53	[G1]
M16 grade 8.8 bolt	94 kN proof load	21	[G2]
Post wall under the 50 mm washer	66 kN punching	15	[G2]
Post, 60 x 60 x 3 at 250 mm	91 MPa, 0.39 of yield	2.6	[G3]
Standard footing 500 x 1,200, drained soil	16.1 kN ultimate	3.58	[G4]
Standard footing 500 x 1,200, saturated soil	7.2 kN ultimate	1.59	[G5]
Earlier footing 400 x 1,000, saturated soil (for comparison)	3.8 kN ultimate	0.85	[G5]

Since Amish's decision 9A of 2026-10-03 the 500 x 1,200 mm footing, with a 2,300 mm post set 1,100 mm into it, is the standard footing at every site. It takes about 0.23 m3 of concrete, about 22 bags of 25 kg [G6]. **R6 is met on paper in drained and in saturated granular soil.** Soft clay or peat banks are outside these screening values and need a site pull test.

H. Deployment, storage and safety (R7, R8, R10)

Lifting the lid (2 s), taking the throw weight and the top of the folded ladder (4 s), throwing upstream (4 s) and letting the ladder pay out (5 s) add up to about 15 s, inside the 30 s of R7; this is an estimate, and R7 can only be closed by timed trials [H1]. The ropes, rungs and foam live in an opaque UV-stabilised box and see daylight only when used or inspected, which is how R8 is met by design; an ageing test closes it [H2]. Nothing thrown is hard: the throw weight is a soft shot pouch, rung ends are capped and holes bushed, and a blunt-tip knife is kept in the lid (R10, met by design) [H3].

I. Cost (R9)

Value-engineering target: USD 500. Estimated cost of the constructable design: USD 605.20 (USD 105.20 over the target) [I1]. The station (post, footing, anchor, box, decals, knife and consumables) is USD 358.80 and the ladder USD 246.40 [I2]. Against the previous estimate, the larger standard footing adds USD 65.00 (22 bags instead of 12 at USD 6.50), the longer post USD 4.60, and the 9 mm ropes save USD 8.00.

Results against the requirements

Table 4. Requirement status at TRL 3.

ID	Status	Figures
R1	Met on paper	4,020 mm rung section; 1.5:1 slopes to 2.1 m, vertical walls to 3.8 m
R2	Met on paper on mass (restated 2026-10-03); throw accuracy needs trials	4.27 kg against 4.5 kg
R3	Met on paper	0.28 and 0.22 kg per rung; bottom three rungs 1.75 times the weight
R4	Met on paper	Rung 0.56 of yield; ropes 18.4 kN wet with knots (4.08 times 4.5 kN)
R5	Met by design; climb time needs trials	298 mm clear, 335 mm pitch, 70 mm hand room
R6	Met on paper in drained and saturated soil	Standard footing 3.58 drained, 1.59 saturated; eye nut 1.53
R7	Not verifiable at TRL 3	Estimate 15 s
R8	Met by design; to confirm by test	Stored in an opaque box
R9	Over the value-engineering target	USD 605.20 against USD 500
R10	Met by design	Soft weight, capped ends, knife

Requirement identifiers (kit 1.9.1)

CNR-PRS-001 now names the requirements PR-001 to PR-013. The notes above use the earlier R1 to R10, which are PR-001 to PR-010 in the same order, and the tags [A1] to [I2] are unchanged. PR-011 (built and mended locally from stock sections and hand tools, no welding for option A) is evidenced by the bill of materials `bom/bom.csv`, whose 24 lines are catalogue stock and hand-cut parts, and by the build plan BLD-001. PR-012 (pictogram how-to, bright box, numbered tamper tag) is evidenced by BOM lines 8 and 11. PR-013 (honest labelling) is checked by inspection of every drawing, render and the README, and by `python .kit/render.py --check`. None of the three needs a calculation.

The independent review of the STEM check (CNR-STM-001, Independent review) raised points this note does not yet cover: the slope-crest effect on the footing of G4 and G5, the eye nut of G1 loaded 14 to 34 degrees off its axis, the rope of E1 after the ageing allowance of R8 (14.7 kN, factor 3.3), and bush bearing with a foot at one rung end in water-saturated nylon (D3, 32.8 MPa against about 35 MPa). Proposed fixes are open decisions 3 to 5 in CNR-DEC-001; no number in this note was changed.